LEARNING MODULE

15

Distributions in Finance

Binomial, Poisson, normal, lognormal and Student-t, each fitted to TAIEX and asked which question it can answer.

VOLUME III · Probability | BUILDS ON LM1 · LM2 · LM3 · LM5 · LM13 · LM14

LEADS TO LM16 · LM17 · LM18 · LM19 · LM20 · LM21 · LM26 · LM32 · LM38 · LM39 | TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 calculate binomial probabilities, means and variances, and compare the binomial with up days in blocks of sessions
☐	2 derive the Poisson distribution as the limit of rare binomial trials, and recognize overdispersion in counts of large moves
☐	3 calculate normal quantiles with Φ^{-1}, and read a QQ plot of returns against a fitted normal
☐	4 derive the density, median, mean and variance of a lognormal growth factor, and fit one to annual growth factors
☐	5 state the Student-t density, variance and kurtosis with their conditions, and fit a Student-t by matching kurtosis and by maximum likelihood
☐	6 compare tail models through exceedance counts and quantiles, and read a tail index from log–log chords
☐	7 choose a distribution for a stated question and name the assumption that limits it
☐	8 implement the fits and counts with `scipy.stats` and the counting guard

LEARNING MODULE OVERVIEW

- TAIEX's up days in 292 blocks of 20 sessions average 10.96 with variance 4.72. A binomial with $\hat{p} = 0.548$ gives 10.96 and 4.95 by (15.2): a variance near $np(1-p)$ is consistent with independent days, not proof of it.
- Rare binomial trials tend to the Poisson, (15.4), whose mean and variance are both λ. Days with a move of 3% or more average 0.612 a month, but their variance is 1.846: the counts are overdispersed, and nine months had five or more against 0.123 expected.
- A normal quantile is $\mu + \sigma\Phi^{-1}(u)$, (15.6), and 1.96 is $\Phi^{-1}(0.975)$. TAIEX's 1% quantile, −3.567%, lies 0.82 points below the normal's −2.749%, and its QQ plot bends away from the line in both tails.
- A lognormal fitted to the 23 calendar-year growth factors 2003–2025 has median 1.1235 (the compound rate, +12.35% a year), mean 1.1516 and a 30.0% chance of a losing year; a model of independent days gives about 26.5%.
- The Student-t has variance $\nu/(\nu-2)$ for $\nu > 2$ and kurtosis $3 + 6/(\nu-4)$ for $\nu > 4$, (15.10). Matching TAIEX's excess kurtosis 5.27 gives $\nu = 5.14$; maximum likelihood gives $\nu = 3.065$, where the kurtosis is infinite.
- At or below −5% the normal expects 0.0817 days, the maximum-likelihood Student-t 19.6, and TAIEX had 15. At −10% the Student-t expects 2.57 days, which the 10% price limit almost excludes.
- One day, 7 April 2025, lifted TAIEX's excess kurtosis from 4.25 to 4.94 and moved its 1% quantile by 0.005 points. Counts and quantiles need fewer moments and moved less; the chord from −5% to −6% rests on a few days and fell from 6.03 to 5.24.
- Each question gets its own model and limiting assumption: the binomial for up days (independence), the lognormal for a year (independent years), the maximum-likelihood Student-t from −3% to −5% (symmetry), and plain counts beyond −7% (the price limit).

1 INTRODUCTION

LM5 fitted a normal model to 0050's daily returns and found its tail far too thin. The model expected 0.058 days at or below −5% in twelve years, and 9 occurred. LM13 and LM14 built the language of random variables, distribution functions and moments. This module applies it to the distributions that finance uses most. Each has one to three parameters and answers some questions well and others badly; the data decide which.

MOTIVATING CASE **Which model for which question?**

Take TAIEX's 5,849 daily total returns from 3 January 2003 to 8 October 2026. A normal model with their mean, 0.0625%, and standard deviation, 1.2084%, expects **0.08** days at or below −5%. TAIEX had **15** (LM13's count). A Student-t fitted by maximum likelihood, with $\nu = 3.065$ degrees of freedom, expects **19.6** (Exhibit 1).

Now count the days with a move of 3% or more, either way, month by month. They average **0.612** a month, but their variance is **1.846**, three times the mean. Three months had eight: September and October 2008, and March 2020.

Which model should answer which question? And what does the crowding of large moves into a few months say about independence?

Exhibit 1: Two models against TAIEX: daily tails and monthly counts

Days with $R \leq -x$	Normal	Student-t, maximum likelihood	TAIEX
$x = 2\%$	257	210	252
$x = 5\%$	0.0817	19.6	15
Months: days with $\lvert R \rvert \geq 3\%$		**Poisson**	**TAIEX**
Mean count		0.612	0.612
Variance of the count		0.612	1.846
Months with five or more		0.123	9

Note: daily simple total returns $R_t = P_t / P_{t-1} - 1$, 3 January 2003 – 8 October 2026, $T = 5{,}849$. Model columns are expected counts. The normal has $\bar{R} = 0.0625\%$ and $s = 1.20836\%$ (divisor T); the Student-t is Section 6's maximum-likelihood fit ($\nu = 3.065$, location 0.0997%, scale 0.7761%). The 286 calendar months run from January 2003 (18 sessions) to October 2026 (6 sessions), and the Poisson has their mean count, $\hat{\lambda} = 175/286$. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm15_numbers.py` and `LM15_lab.ipynb`.

Sections 2 and 3 count events with the binomial and the Poisson. Sections 4 to 6 fit the normal, the lognormal and the Student-t. Section 7 compares tail measures and chooses a model per question, and Section 8 collects the Python calls.

NOTE **Notation for this module**

The sample is TR23, TAIEX's total-return index (cash dividends reinvested): $T = 5{,}849$ simple daily returns $R_t = P_t/P_{t-1} - 1$ from 3 January 2003 to 8 October 2026. The closes, 2 January 2003 – 8 October 2026, span 23.7645 years, so $A = 246.12$ sessions a year. Moments divide by T: $\bar{R}$ and s for R, $\bar{r}$ and s_r for $r = \ln(1+R)$. Counts use the guard $R \le -x + 10^{-12}$, because two-decimal index levels make round returns exact ties, and an empirical quantile is LM14's order statistic $\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$.

Local symbols: X a generic random variable, per section; $Y \sim t_\nu$; y a generic coordinate in (15.7)–(15.9) and Deep Dive 2 (no yields here); $a > 0$ the argument of Γ (no bets here); $\hat{p}$ the blocks' up-day frequency; $\hat{\lambda}$ the months' mean count of large moves; ρ_1 the correlation of adjacent up-day indicators; G a year's growth factor, with μ_X, σ_X the mean and standard deviation of $X = \ln G$; $m = 246$ the sessions of a model year; $x > 0$ a loss level, $x_1 < x_2$ two of them; α a tail index (LM5).

NOTE **Research hygiene**

Every frequency here is **descriptive** of TAIEX's total returns, 3 January 2003 – 8 October 2026. Every fitted distribution is **in-sample**, so a good fit is no evidence for later data. Comparing the 7% and 10% limit regimes is a **stability check**, and the Poisson limit and LM17's one-year probability are **model** statements. LM20 and LM21 add standard errors and tests.

2 COUNTING SUCCESSES: BERNOULLI AND BINOMIAL

- ☐ calculate binomial probabilities, means and variances, and compare the binomial with up days in blocks of sessions

TR23's first 20 returns, 3 January to 7 February 2003, contain 12 up days. Is 12 many? The answer needs a model for the count.

A **Bernoulli trial** has two outcomes: a success with probability p and a failure with probability $1-p$. Its indicator $X = \mathbf{1}\{\text{success}\}$ takes the values 1 and 0, so (14.1) gives $\mathbb{E}[X] = p$. Because $X^2 = X$, also $\mathbb{E}[X^2] = p$, and (14.4) gives $\operatorname{Var}(X) = p - p^2 = p(1-p)$.

DEFINITION **Binomial distribution**

Let n independent trials ((13.13) of LM13) each succeed with the same probability p, $0 < p < 1$. The number of successes X has the **binomial distribution** $\operatorname{Bin}(n, p)$:

$$\Pr(X = j) = \binom{n}{j} p^j (1-p)^{n-j}, \qquad j = 0, 1, \ldots, n. \tag{15.1}$$

By (13.7), one given sequence with j successes has probability $p^j(1-p)^{n-j}$. There are $\binom{n}{j}$ such sequences, one for each choice of the positions of the successes, and they are disjoint, so their probabilities add. By the binomial theorem of LM1's Deep Dive on e, the probabilities (15.1) sum to $(p+1-p)^n = 1$.

For the moments, use $j\binom{n}{j} = n\binom{n-1}{j-1}$ and $j(j-1)\binom{n}{j} = n(n-1)\binom{n-2}{j-2}$, and then the binomial theorem again:

$$\mathbb{E}[X] = np\sum_{j=1}^{n}\binom{n-1}{j-1}p^{j-1}(1-p)^{n-j} = np,$$
$$\mathbb{E}[X(X-1)] = n(n-1)p^2.$$

Since $\operatorname{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - (\mathbb{E}[X])^2$ by (14.4),

$$\begin{aligned}\mathbb{E}[X] &= np,\\ \operatorname{Var}(X) &= n(n-1)p^2 + np - n^2p^2\\ &= np(1-p).\end{aligned} \tag{15.2}$$

LM16 reaches this variance in one line, adding the variances of independent indicators. LM17 shows why a sum of many independent trials is close to normal.

EXAMPLE 1

Up days in a block

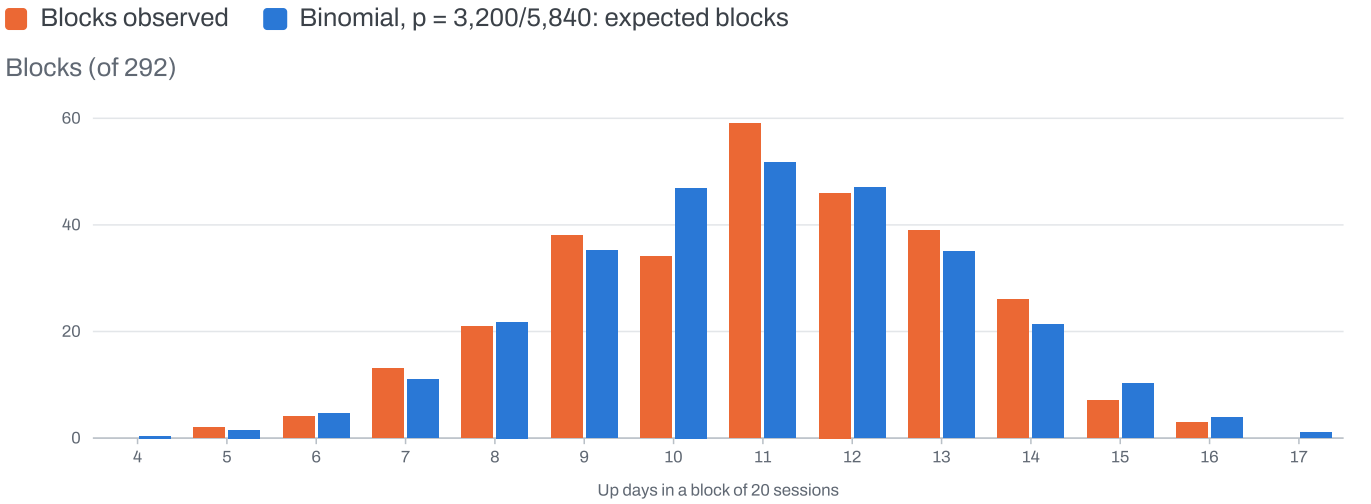
TR23's first 5,840 returns form 292 blocks of 20 sessions; the last 9 returns, 24 September – 8 October 2026, are excluded. The blocks contain 3,200 up days (no return is exactly 0), so their up-day frequency is $\hat{p} = 3{,}200/5{,}840 = 0.548$. Model each block's count as $\operatorname{Bin}(20, 0.548)$. Compute its mean and variance and the expected numbers of blocks with at most 5 and at least 16 up days.

SOLUTION

By (15.2), the mean is $20 \times 0.548 = 10.96$ and the variance $20 \times 0.548 \times 0.452 = 4.95$. The first block's 12 up days lie within one standard deviation, 2.23, of the mean 10.96. The 292 blocks have mean 10.96, by construction, and variance 4.72 (divisor 292).

By (15.1), $\Pr(X \le 5) = 0.00677$, or $292 \times 0.00677 = 1.98$ blocks, against 2 observed. Similarly $\Pr(X \ge 16) = 0.01807$, or 5.28 blocks, against 3. Exhibit 2 compares every count.

Exhibit 2: Up days in 292 blocks of 20 sessions against the binomial



Note: TR23's first 5,840 returns, 3 January 2003 – 23 September 2026, in consecutive blocks of 20; an up day has $R_t > 0$. Expected blocks are $292\,\Pr(X = j)$ under $\mathrm{Bin}(20, 3{,}200/5{,}840)$, from the unrounded $\hat{p}$. Outside 4 to 17 up days the binomial expects 0.34 blocks. Source: as Exhibit 1.

PITFALL **Independent trials with one p**

A variance near $np(1-p)$ is consistent with independent days; it does not prove independence. In LM13's 5,848 pairs of adjacent days, the second day was down with frequency 0.4521, and with 0.4639 after a down day. Let X_1, X_2 be a pair's down-day indicators, both taken at frequency 0.4521 (the first day's is 0.4519). Their covariance $\mathbb{E}[X_1X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2]$ (LM16) is $0.4521 \times (0.4639 - 0.4521)$ and each variance $0.4521 \times (1 - 0.4521)$, so their correlation is $\rho_1 = (0.4639 - 0.4521)/(1 - 0.4521) = 0.0215$ (0.0214 from the unrounded frequencies), as for up-day indicators.

If only adjacent days are correlated, LM16's variance of a sum carries ρ_1 into a block of 20:

$$20 \times 0.548 \times 0.452 \times \left(1 + 2 \cdot \tfrac{19}{20} \cdot 0.0215\right) = 5.16.$$

The blocks give 4.72, below both 5.16 and 4.95: 292 blocks cannot see a dependence this small either way (LM21 tests such gaps).

KNOWLEDGE CHECK 1

For $n = 20$ trials, which p maximizes the binomial variance? Compare the blocks' 4.95 with that maximum.

ANSWER

The variance $np(1-p)$ is a downward parabola in p with derivative $n(1-2p)$, zero at $p = \frac{1}{2}$. Its maximum is $n/4 = 5$. The blocks' $\hat{p} = 0.548$ lies near $\frac{1}{2}$, so 4.95 lies near 5: there the variance hardly depends on p.

3 RARE EVENTS: THE POISSON DISTRIBUTION

- ☐ derive the Poisson distribution as the limit of rare binomial trials, and recognize overdispersion in counts of large moves

TR23 has 175 days with $|R| \geq 3\%$ among 5,849 (LM14); set p to the frequency $175/5{,}849 = 0.02992$ a session. A month has about 20 sessions, so its count of large moves looks like a binomial with many trials and a small p. What matters is the mean $\lambda = np$. Fix λ and let n grow. At $j = 0$, (1.6) with $-\lambda$ in place of its variable gives

$$\Pr(X = 0) = \left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}.$$

DEFINITION **Poisson distribution**

A count X has the **Poisson distribution** $\text{Poisson}(\lambda)$, with $\lambda > 0$, when

$$\Pr(X = j) = \frac{e^{-\lambda}\lambda^j}{j!}, \qquad j = 0, 1, 2, \ldots \tag{15.3}$$

THEOREM **Poisson limit of the binomial**

Fix $\lambda > 0$ and a count $j \geq 0$. Then, as $n \to \infty$,

$$\binom{n}{j}\left(\frac{\lambda}{n}\right)^j\left(1 - \frac{\lambda}{n}\right)^{n-j} \to \frac{e^{-\lambda}\lambda^j}{j!}. \tag{15.4}$$

Deep Dive 1 proves (15.4) for every j. By (3.2), the probabilities (15.3) sum to $e^{-\lambda}e^{\lambda} = 1$. The mean comes from the same series, after the term $j = 0$ is dropped and the index shifted:

$$\mathbb{E}[X] = \sum_{j=1}^{\infty} j\,\frac{e^{-\lambda}\lambda^{j}}{j!} = \lambda \sum_{j=1}^{\infty} \frac{e^{-\lambda}\lambda^{j-1}}{(j-1)!} = \lambda.$$

The same shift gives $\mathbb{E}[X(X-1)] = \lambda^2$ (Problem 4), so

$$\mathbb{E}[X] = \mathrm{Var}(X) = \lambda. \tag{15.5}$$

A Poisson count has a variance equal to its mean, the property that monthly counts of large moves fail in Example 3.

DEEP DIVE **Deep Dive 1: why the binomial tends to the Poisson** optional · click to expand

Fix j and take $n > \lambda$. Split the binomial probability into four factors:

$$\begin{aligned}\binom{n}{j}\left(\frac{\lambda}{n}\right)^{j}\left(1-\frac{\lambda}{n}\right)^{n-j} &= \frac{\lambda^{j}}{j!}\cdot\frac{n(n-1)\cdots(n-j+1)}{n^{j}} \\ &\quad\cdot\left(1-\frac{\lambda}{n}\right)^{n}\cdot\left(1-\frac{\lambda}{n}\right)^{-j}.\end{aligned}$$

The second factor is a product of j terms $(n-l)/n$, $l < j$, each tending to 1, as in LM1's Deep Dive on e. The third tends to $e^{-\lambda}$ by (1.6). The fourth is a fixed power of a term that tends to 1. So the product tends to $e^{-\lambda}\lambda^{j}/j!$. ■

EXAMPLE 2

From a rare binomial to the Poisson

A 20-session month in which each session, independently, has a large move with probability 0.02992 has a count $X \sim \mathrm{Bin}(20, 0.02992)$, with mean $\lambda = 0.5984$. Compare its pmf with that of $\mathrm{Poisson}(0.5984)$ at $j = 0$, 1 and 2. Then take the case's monthly mean, 0.612 ($\hat{\lambda}$ of Example 3), and compute $(1 - 0.612/n)^{n}$ for $n = 20$, 250 and 10,000, against $e^{-0.612}$.

SOLUTION

By (15.1) and (15.3):

j	$\text{Bin}(20, 0.02992)$	$\text{Poisson}(0.5984)$
0	0.5447	0.5497
1	0.3360	0.3289
2	0.09845	0.09842

The two pmfs differ by at most 0.0071 here. The limit at $j = 0$ is slow: $(1 - 0.612/n)^n$ is 0.53711, 0.54186 and 0.54226 for $n = 20$, 250 and 10,000, against $e^{-0.612} = 0.54227$. The gap, 0.0052 at $n = 20$ and 0.0004 at $n = 250$, is close to $e^{-\lambda}\lambda^2/(2n)$ with $\lambda = 0.612$ (0.0051 and 0.0004). The reason is that, by (3.3),

$$n \ln\left(1 - \frac{\lambda}{n}\right) = -\lambda - \frac{\lambda^2}{2n} - \dots,$$

and the error of the approximation is of order $1/n^2$.

EXAMPLE 3

Do large moves arrive like a Poisson count?

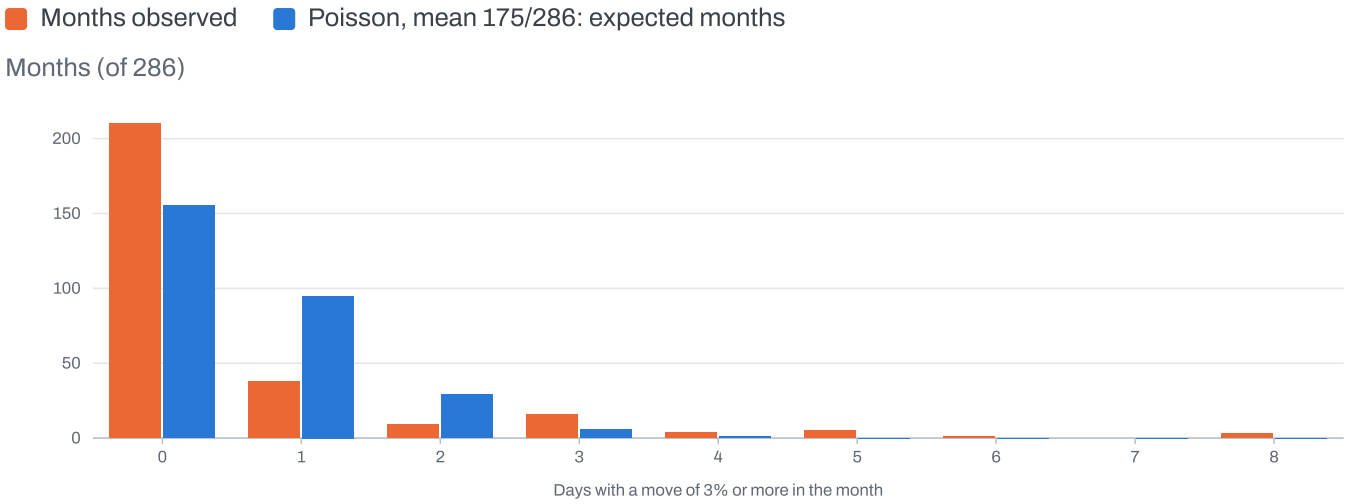
TR23's 286 calendar months, January 2003 (18 sessions) to October 2026 (6 sessions), contain the 175 days with $|R| \geq 3\%$. Fit a Poisson by the months' mean count, and compare expected and observed months.

SOLUTION

The fitted mean is $\hat{\lambda} = 175/286 = 0.612$. By (15.3), 155.1 of the 286 months should have no large move, and 210 had none. One large move: 94.9 expected, 38 observed. Five or more: 0.123 against 9. Eight or more: 8.1×10^{-5} against 3 (Exhibit 3).

The counts have variance 1.846 (divisor 286), 3.02 times their mean, where (15.5) requires 1: they are **overdispersed**. Months of unequal length (6 to 24 sessions) do not explain it. With one p for every session, the months' binomial variances average 0.594, and their means have variance 0.0055. By LM16's law of total variance, the count's variance is their sum, only 0.60. Each limit regime alone has a ratio near 3: 2.95 before June 2015, 3.01 after.

Exhibit 3: Months by their number of large moves: TAIEX against the Poisson



Note: days with $|R| \geq 3\%$ (guarded) in each of the 286 calendar months, January 2003 – October 2026. Expected months under $\text{Poisson}(175/286)$, for 0 to 8 large moves: 155.1, 94.9, 29.0, 5.92, 0.906, 0.111, 0.0113, 9.9×10^{-4} and 7.6×10^{-5}. Eight or more: 8.1×10^{-5} expected, 3 observed (September and October 2008, March 2020); December 2008 had six. Source: as Exhibit 1.

Calm and turbulent months come in runs because volatility changes over time; LM26 models it. LM39 uses Poisson counts for price jumps. Grinstead and Snell (1997, §5.1, Table 5.1) tabulate the binomial against its Poisson limit.

PITFALL **Rare is not independent**

A Poisson count needs rare events that arrive independently, at a constant rate. Large moves are rare, but they cluster. The fitted Poisson matches the average month and misses the bad ones: 9 months with five or more large moves against 0.123 expected. Check a count's variance before modelling it as a Poisson.

導讀 **二項分布與卜瓦松分布：計數的模型**

一個交易日漲或不漲，可以用柏努利分布（Bernoulli distribution）描述：成功機率 p，平均數（mean）p，變異數（variance）$p(1-p)$。n 次獨立（independent）而成功機率相同的試驗，成功次數服從二項分布（binomial distribution），平均數 np、變異數 $np(1-p)$。加權股價報酬指數每 20 個交易日一組，上漲天數的變異數是 4.72，與二項分布的 4.95 相近，但這只表示資料與獨立相容，並不是證明。當 p 很小、n 很大而 $np = \lambda$ 固定時，二項分布趨近卜瓦松分布（Poisson distribution），它的平均數與變異數都等於 λ。這個指數每月單日漲跌 3% 以上的天數平均 0.612 天，變異數卻是 1.846，約為平均數的三倍，這叫過度離散（overdispersion）：大波動會群聚，不像獨立同分布（iid）的稀有事件（rare event）。

KNOWLEDGE CHECK 2

Why not model the up days of a 20-session block with a Poisson?

ANSWER

The Poisson is the limit of a binomial with a small p, and up days have $\hat{p} = 0.548$. A Poisson's variance equals its mean, while a binomial's is $1 - p$ times its mean: 4.95 against 10.96 here.

4 THE NORMAL DISTRIBUTION: QUANTILES AND QQ PLOTS

- [ ] calculate normal quantiles with Φ^{-1}, and read a QQ plot of returns against a fitted normal

LM5 computed normal probabilities, $\Pr(R \le -x) = \Phi\big((-x-\mu)/\sigma\big)$. Risk questions often run the other way: which loss is reached on 1% of days? That loss is a quantile. LM14's quantile function $F_X^{-1}(u)$ is the smallest value at which F_X reaches u. Because Φ is continuous and strictly increasing from 0 to 1, exactly one z solves $\Phi(z) = u$ for each u in $(0,1)$; that z is $\Phi^{-1}(u)$.

PROPOSITION Normal quantiles

Let $R \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$, and let $0 < u < 1$. Then

$$\begin{aligned} F_R^{-1}(u) &= \mu + \sigma\,\Phi^{-1}(u), \\ \Phi^{-1}(1-u) &= -\Phi^{-1}(u). \end{aligned} \tag{15.6}$$

Proof. By (5.5), $\Pr\big(R \le \mu + \sigma\Phi^{-1}(u)\big) = \Phi\big(\Phi^{-1}(u)\big) = u$, and F_R is strictly increasing. The symmetry $\Phi(-z) = 1 - \Phi(z)$ of LM5 gives $\Phi\big(-\Phi^{-1}(u)\big) = 1 - u$. ■

LM4 used 1.96, without proof, as the value that a standard normal variable exceeds with probability 2.5%. By (15.6) it is $\Phi^{-1}(0.975) = -\Phi^{-1}(0.025) = 1.9600$. With its sign reversed, the 1% quantile of a daily return is the one-day value at risk at 99% (LM32).

EXAMPLE 4

Normal quantiles against TAIEX's

SciPy's `norm.ppf` gives Φ^{-1} at 0.95, 0.975, 0.99 and 0.999 as 1.6449, 1.9600, 2.3263 and 3.0902. With $\bar{R} = 0.0625\%$ and $s = 1.2084\%$, compute the normal's 5%, 2.5%, 1% and 0.1% quantiles. Compare them with TAIEX's empirical quantiles $\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$ (LM14).

SOLUTION

By (15.6) and symmetry, the normal's 5% quantile is $0.0625\% - 1.2084\% \times 1.6449 = -1.925\%$; the other levels follow in the same way (three decimals):

Level u	Rank $\lceil uT \rceil$	Normal (%)	TAIEX (%)
5%	293	−1.925	−1.862
2.5%	147	−2.306	−2.648
1%	59	−2.749	−3.567
0.1%	6	−3.672	−6.313

At 5% the normal is slightly too deep. From 2.5% on it is too shallow, and the gap widens to 0.82 points at 1% and 2.64 points at 0.1%. TAIEX's 0.1% quantile is its sixth-worst day, 11 October 2018.

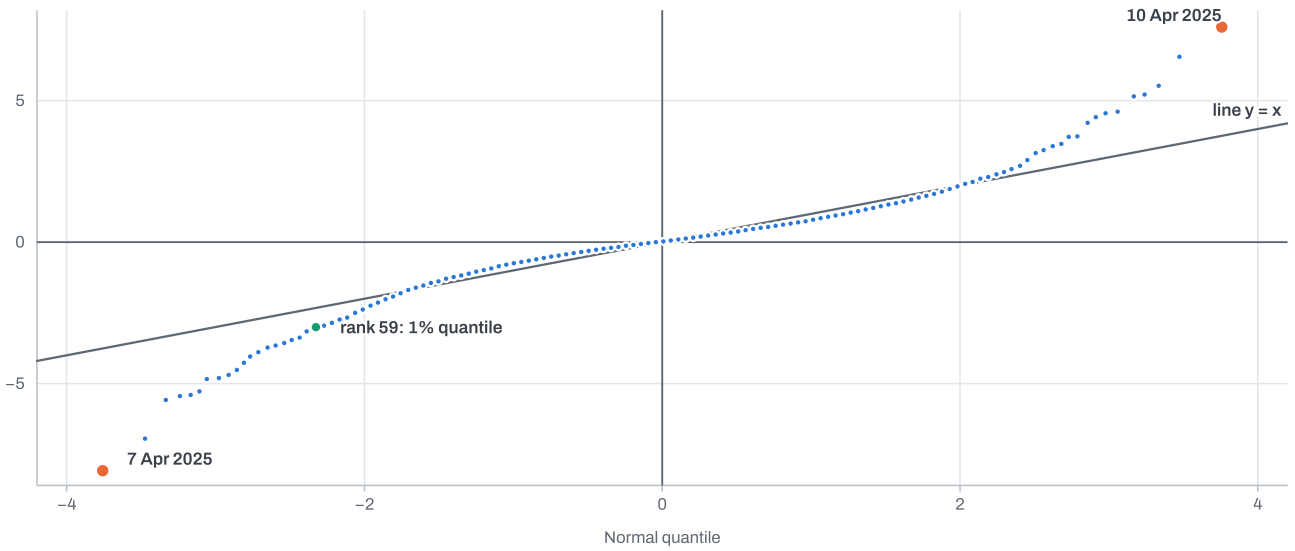
DEFINITION **QQ plot**

Sort the T returns, $R_{(1)} \le \cdots \le R_{(T)}$. The **QQ plot** (quantile–quantile plot) of the returns against a fitted normal places rank j at the point

$$\left(\Phi^{-1}\!\left(\frac{j - \tfrac{1}{2}}{T} \right),\ \frac{R_{(j)} - \bar{R}}{s} \right), \qquad j = 1, \dots, T.$$

The coordinates are the normal quantile and the standardized return; normal returns lie near the 45° line.

Exhibit 4: QQ plot of TAIEX's daily returns against the fitted normal



Note: 130 of the 5,849 points, standardized with $\bar{R} = 0.0625\%$ and $s = 1.20836\%$. Plotted are ranks 1, 59 and 5,849, and each rank at least 0.05 right of the last plotted point or 0.2 above it. The omitted points hide no cited value. Below the line on the left and above it on the right, the tails are heavier than the normal's. Rank 59, the 1% quantile, sits at $(-2.326, -3.004)$. Source: as Exhibit 1.

EXAMPLE 5

Reading the QQ plot

Locate the worst day, the best day and rank 59 in Exhibit 4, and explain the plot's S shape.

SOLUTION

The worst day, 7 April 2025, sits at $(-3.758, -8.079)$. The lowest of 5,849 normal draws lies about 3.7 standard deviations below the mean; TAIEX's lies 8.08 below. The best day, 10 April 2025, sits at $(3.758, 7.604)$.

Rank 59 sits at $(-2.326, -3.004)$, 0.68 standard deviations below the line: Example 4's 1% gap of 0.82 points, divided by $s = 1.2084\%$. The plot has an S shape, below the line on the left and above it on the right. That shape means more mass in both tails than the normal has: heavy tails.

PITFALL **Which level does a QQ point stand for?**

Rank j sits at the level $(j - \frac{1}{2})/T$, the middle of the step of $\hat{F}$ from $(j-1)/T$ to j/T; at j/T the best day would sit at $\Phi^{-1}(1) = +\infty$. A quantile quoted at a stated level keeps LM14's rule $R_{(\lceil uT \rceil)}$: the QQ convention is for drawing, not for quoting.

導讀 **常態分位數與分位數-分位數點圖**

機率是給定門檻求面積，分位數（quantile）反過來，是給定機率求門檻。常態分布（normal distribution）的分位數函數（quantile function）是 $\mu + \sigma\Phi^{-1}(u)$，LM4 用到的 1.96 就是 $\Phi^{-1}(0.975)$。在 1% 的水準，常態模型給 −2.749%，加權股價報酬指數經驗分布（empirical distribution）的 1% 分位數卻是 −3.567%，相差 0.82 個百分點，而且越往尾端差得越多。分位數-分位數點圖（QQ plot）把排序後的標準化報酬對常態分位數作圖：資料若接近常態，點會落在 45 度線附近；這個指數的點左端低於直線、右端高於直線，呈 S 形，表示兩側尾巴都比常態厚，也就是厚尾（heavy tail）。

KNOWLEDGE CHECK 3

Which standard deviation σ would make a normal with mean 0.0625% expect 15 days at or below −5% in 5,849? How many days at or below −2% would it then expect?

ANSWER

The model needs $5{,}849\,\Phi\big((-5\% - 0.0625\%)/\sigma\big) = 15$, so $(-5.0625\%)/\sigma = \Phi^{-1}(15/5{,}849) = -2.7988$ and $\sigma = 1.809\%$. That normal expects $5{,}849\,\Phi(-2.0625/1.809) = 743$ days at or below −2%, against 252. A wider bell moves the whole curve; the data need a different shape.

5 THE LOGNORMAL DISTRIBUTION

- [ ] derive the density, median, mean and variance of a lognormal growth factor, and fit one to annual growth factors

A year's growth factor G is positive, a product of many daily factors, so its logarithm $X = \ln G$ is a sum. LM17 explains why a product of many independent factors is close to lognormal; LM38 builds lognormal prices.

DEFINITION **Lognormal distribution**

Let $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ with $\sigma_X > 0$. Then $G = e^X$ has the **lognormal distribution** with parameters μ_X and σ_X.

PROPOSITION **Density, quantiles and moments of a lognormal**

Let $G = e^X$ with $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $\sigma_X > 0$. Then, for $y > 0$ and $0 < u < 1$,

$$\begin{aligned} f_G(y) &= \frac{1}{\sigma_X y}\,\varphi\Big(\frac{\ln y - \mu_X}{\sigma_X}\Big), \\ F_G^{-1}(u) &= e^{\mu_X + \sigma_X \Phi^{-1}(u)}, \\ \mathbb{E}[G] &= e^{\mu_X + \sigma_X^2/2}, \\ \operatorname{Var}(G) &= \left(e^{\sigma_X^2} - 1\right) e^{2\mu_X + \sigma_X^2}. \end{aligned} \tag{15.7}$$

In particular the median is e^{μ_X}, below the mean.

Proof. The exponential is increasing, so $F_G(y) = \Pr(X \le \ln y) = \Phi\big((\ln y - \mu_X)/\sigma_X\big)$ for $y > 0$. The chain rule (2.5) and $\Phi' = \varphi$ (LM5) give the density. The same monotonicity and (15.6) give the quantiles; the median, at $u = \frac{1}{2}$, is e^{μ_X}. The mean is (5.12). For the variance, $2X \sim \mathcal{N}(2\mu_X, 4\sigma_X^2)$, so (5.12) gives $\mathbb{E}[G^2] = e^{2\mu_X + 2\sigma_X^2}$; subtract $(\mathbb{E}[G])^2 = e^{2\mu_X + \sigma_X^2}$, by (14.4). ■

EXAMPLE 6

A lognormal for annual growth

TR23 has 23 calendar-year growth factors G, 2003–2025, with 2003 measured from its first close (2 January 2003). Their logarithms have mean $\mu_X = 0.11646$ and standard deviation $\sigma_X = 0.22207$ (divisor 23). LM14 found an arithmetic mean return of 14.97% and a compound rate of 12.35%. Fit a lognormal with these parameters. Compute its median, mean and standard deviation, the probability of a losing year, and the 5% and 95% quantiles.

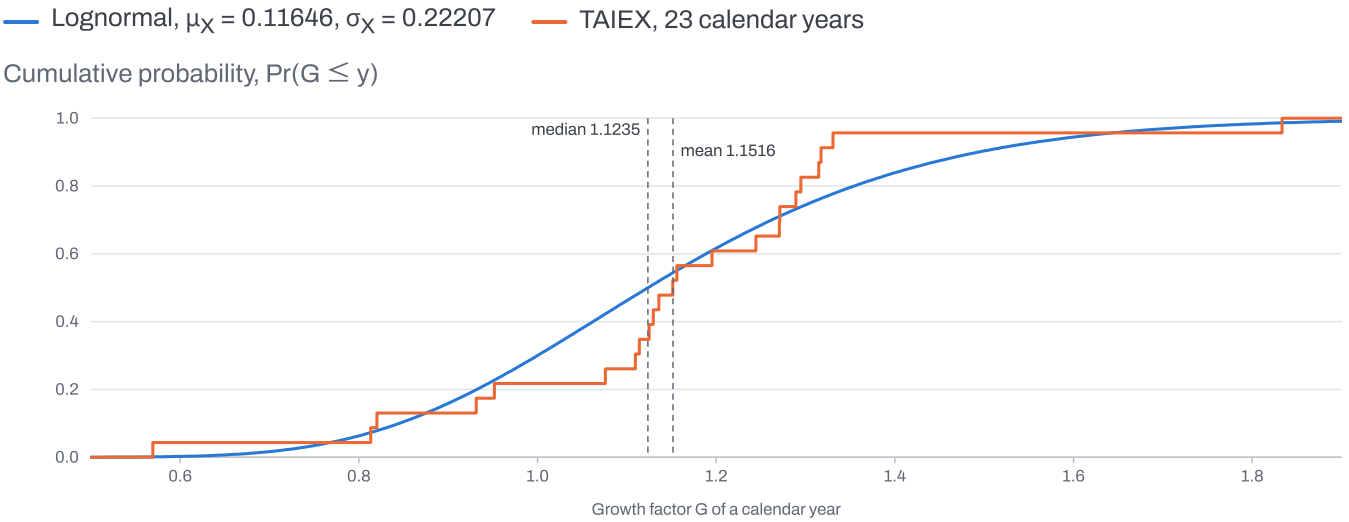
SOLUTION

By (15.7), the median is $e^{0.11646} = 1.1235$, or +12.35%: the compound rate, because e^{μ_X} is the geometric mean of the factors. The mean is $e^{0.11646 + 0.22207^2/2} = 1.1516$, +15.16% against the sample's 14.97%. The standard deviation is 0.2589.

A losing year has probability $\Pr(G < 1) = \Phi(-0.11646/0.22207) = \Phi(-0.5244) = 30.0\%$: 6.90 of 23 years, against 5 observed (2008, 2011, 2015, 2018 and 2022). The 5% and 95% quantiles are $e^{0.11646 \mp 1.6449 \times 0.22207}$, that is 0.7797 and 1.6189 (Exhibit 5).

LM17 models a year instead as $m = 246$ independent daily log returns, fitted to the 5,664 log returns from 3 January 2003 to 31 December 2025. Its annual log return has mean 0.1163, close to μ_X, but standard deviation 0.1856 against $\sigma_X = 0.22207$. By LM17's normal approximation, a losing year then has probability 26.5%. Independent days scaled by $\sqrt{m}$ understate the annual spread (LM19).

Exhibit 5: The lognormal fitted to 23 calendar years of TAIEX



Note: distribution functions of the growth factor G. The lognormal has $\mu_X = 0.11646$ and $\sigma_X = 0.22207$; TAIEX's 23 calendar-year factors (2003 from its first close) each add a step of $1/23$. Vertical lines: the model's median 1.1235 and mean 1.1516. Beyond the model's 5% and 95% quantiles, 0.7797 and 1.6189, lie 2 years (2008, 0.5693; 2009, 1.8334), against 2.3 expected. The empirical median is 1.1514 (2013). Source: as Exhibit 1.

PITFALL **A lognormal does not fix daily tails**

A lognormal for $1 + R$ is a normal for the log return r. With $\bar{r} = 0.05517\%$ and $s_r = 1.2107\%$, it expects $5{,}849\,\Phi\big((\ln 0.95 - \bar{r})/s_r\big) = 0.0541$ days at or below −5%, against the normal's 0.0817 and TAIEX's 15. Taking logs barely changes −5%: the problem is the shape of the tail, not the variable.

導讀 **對數常態分布：年度成長倍數**

若 $X = \ln G$ 服從常態分布（normal distribution），成長倍數 $G = e^X$ 就服從對數常態分布（lognormal distribution）：G 恆為正，中位數（median）是 e^{μ_X}，平均數（mean）是 $e^{\mu_X + \sigma_X^2/2}$，所以平均數一定高於中位數。以加權股價報酬指數 2003–2025 年 23 個年度成長倍數的對數平均數與標準差（standard deviation）為參數，中位數 1.1235 正好對應年複合報酬率 12.35%，平均數為 1.1516，虧損年的機率是 30.0%，而實際上 23 年中有 5 年虧損。若改用 LM17 的獨立同分布（iid）日報酬模型，年度對數報酬（log return）的標準差只有 0.1856，以常態逼近（normal approximation）計算，虧損年機率降為約 26.5%。對數常態只是把常態套在對數報酬上，修正不了日報酬的厚尾（heavy tail）。

KNOWLEDGE CHECK 4

A normal fitted to the 23 growth factors themselves has mean 1.1497 and standard deviation 0.2370 (divisor 23). What probability does it give to $G < 0$, a loss of more than the whole investment? What does the lognormal give?

ANSWER

The normal gives $\Pr(G < 0) = \Phi(-1.1497/0.2370) = 6.1 \times 10^{-7}$: tiny, but positive for an outcome that an index without leverage cannot produce. The lognormal gives 0, because $G = e^X > 0$.

6 THE STUDENT-T DISTRIBUTION

- [ ] state the Student-t density, variance and kurtosis with their conditions, and fit a Student-t by matching kurtosis and by maximum likelihood

LM5 wrote the Student-t density with $\nu = 3$ "up to a constant" and quoted its kurtosis $3 + 6/(\nu - 4)$ without proof. This section supplies the constant, proves the moments and fits the distribution to TAIEX. LM14's Chebyshev and Cantelli bounds cap a tail probability but predict no count; a count needs a model with a shape.

The gamma function

DEFINITION **Gamma function**

For $a > 0$,

$$\Gamma(a) := \int_0^\infty y^{a-1} e^{-y}\, dy, \qquad \Gamma(a+1) = a\,\Gamma(a), \qquad \Gamma(\tfrac{1}{2}) = \sqrt{\pi}. \tag{15.8}$$

The integral converges for $a > 0$: near 0 by LM5's power test, and at infinity because an exponential beats any power. Integration by parts (5.6), differentiating y^a and integrating e^{-y}, gives $\Gamma(a+1) = \big[-y^a e^{-y}\big]_0^\infty + a\,\Gamma(a) = a\,\Gamma(a)$. With $\Gamma(1) = 1$, the gamma function extends the factorial: $\Gamma(2) = 1$, $\Gamma(3) = 2$, $\Gamma(4) = 6$. For $\Gamma(\tfrac{1}{2})$, substitute $y = z^2/2$ by (5.4) and use (5.7):

$$\Gamma(\tfrac{1}{2}) = \int_0^\infty \sqrt{2}\, e^{-z^2/2}\, dz = \sqrt{2} \cdot \frac{\sqrt{2\pi}}{2} = \sqrt{\pi}.$$

The density and its moments

DEFINITION **Student-t distribution**

A random variable Y has the standard **Student-t distribution** with $\nu > 0$ **degrees of freedom**, $Y \sim t_\nu$, when

$$f_Y(y) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{y^2}{\nu}\right)^{-\frac{\nu+1}{2}} \tag{15.9}$$

for every real y.

The constant makes the total area 1. It follows from the beta integral and is stated here without proof (SciPy's `t` reference; Blitzstein and Hwang 2019, §10.4, treat the distribution). The numbers script checks the area numerically. The density is symmetric about 0, and its tails fall like $|y|^{-(\nu+1)}$. By LM5's power test, the moments of order below ν exist and the others do not, so the mean, 0, exists for $\nu > 1$.

Two limits frame the family. At $\nu = 1$ the density is $1/\left(\pi(1+y^2)\right)$, the Cauchy law of LM5's tails exhibit, with no mean. As $\nu \to \infty$, $(1+y^2/\nu)^\nu \to e^{y^2}$ by (1.6), so the shape tends to $e^{-y^2/2}$ and t_ν tends to $\mathcal{N}(0,1)$; Example 7 checks the constant numerically.

EXAMPLE 7

Constants from the gamma function

Compute $\Gamma(\frac{1}{2})$, $\Gamma(\frac{3}{2})$, $\Gamma(2)$ and $\Gamma(\frac{5}{2})$. Then evaluate the constant of (15.9) at $\nu = 3$, 1, 30 and 1,000.

SOLUTION

By (15.8), $\Gamma(\frac{1}{2}) = \sqrt{\pi} = 1.772454$, $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi} = 0.886227$, $\Gamma(2) = 1$ and $\Gamma(\frac{5}{2}) = \frac{3}{2}\Gamma(\frac{3}{2}) = 1.329340$.

At $\nu = 3$ the constant is $\Gamma(2)/\left(\sqrt{3\pi}\,\Gamma(\frac{3}{2})\right) = 2/(\pi\sqrt{3}) = 0.3676$, the constant that LM5's text omitted. The density $\frac{2}{\pi\sqrt{3}}(1+y^2/3)^{-2} = 6\sqrt{3}/\left(\pi(3+y^2)^2\right)$ is the t_3 density that LM5's widget draws.

At $\nu = 1$ the constant is $\Gamma(1)/\left(\sqrt{\pi}\,\Gamma(\frac{1}{2})\right) = 1/\pi = 0.3183$, the Cauchy constant. It is 0.3956 at $\nu = 30$ and 0.3988 at $\nu = 1{,}000$, approaching the normal's $1/\sqrt{2\pi} = 0.3989$.

THEOREM **Moments of the Student-t**

Let $Y \sim t_\nu$. Then

$$\begin{aligned} \operatorname{Var}(Y) &= \frac{\nu}{\nu-2} \quad (\nu > 2), \\ \operatorname{Kurt}(Y) &= 3 + \frac{6}{\nu-4} \quad (\nu > 4). \end{aligned} \tag{15.10}$$

For $\nu \le 2$ the second moment is infinite. For $2 < \nu \le 4$ the kurtosis is infinite, and for $\nu \le 2$ it is undefined.

Deep Dive 2 derives (15.10). To give a return model a chosen mean μ and standard deviation σ, scale Y:

$$R = \mu + \sigma\sqrt{\frac{\nu-2}{\nu}}\, Y, \qquad \nu > 2. \tag{15.11}$$

By (14.3), (14.4) and (15.10), this R has mean μ and variance $\sigma^2 \cdot \frac{\nu-2}{\nu} \cdot \frac{\nu}{\nu-2} = \sigma^2$. A shift and a positive scale leave the kurtosis unchanged, so matching a kurtosis $\operatorname{Kurt} > 3$ means solving $3 + 6/(\nu-4) = \operatorname{Kurt}$:

$$\nu = 4 + \frac{6}{\operatorname{Kurt} - 3}. \tag{15.12}$$

DEEP DIVE **Deep Dive 2: the variance and kurtosis of the Student-t** optional · click to expand

Here y is the variable of f_Y and z a rescaled variable. By the chain rule (2.5),

$$y\Big(1+\frac{y^2}{\nu}\Big)^{-\frac{\nu+1}{2}} = -\frac{\nu}{\nu-1}\,\frac{d}{dy}\Big(1+\frac{y^2}{\nu}\Big)^{-\frac{\nu-1}{2}}.$$

Integrate y times the left side by parts (5.6). For $\nu>2$ the boundary term, of order $|y|^{2-\nu}$, vanishes, and

$$\int_{-\infty}^{\infty} y^2\Big(1+\frac{y^2}{\nu}\Big)^{-\frac{\nu+1}{2}}dy = \frac{\nu}{\nu-1}\int_{-\infty}^{\infty}\Big(1+\frac{y^2}{\nu}\Big)^{-\frac{\nu-1}{2}}dy.$$

Substitute $y=z\sqrt{\nu/(\nu-2)}$ by (5.4). The integrand becomes $\big(1+z^2/(\nu-2)\big)^{-((\nu-2)+1)/2}$, the shape of $t_{\nu-2}$, so the integral on the right is $\sqrt{\nu/(\nu-2)}$ times the reciprocal of (15.9)'s constant at $\nu-2$. Multiply by the constant at ν and use $\Gamma(a+1)=a\Gamma(a)$ twice:

$$\begin{aligned}\mathbb{E}[Y^2] &= \frac{\nu}{\nu-1}\sqrt{\frac{\nu}{\nu-2}}\\ &\quad\times\frac{\Gamma\big(\frac{\nu+1}{2}\big)\,\Gamma\big(\frac{\nu-2}{2}\big)\sqrt{(\nu-2)\pi}}{\Gamma\big(\frac{\nu-1}{2}\big)\,\Gamma\big(\frac{\nu}{2}\big)\sqrt{\nu\pi}}\\ &= \frac{\nu}{\nu-1}\cdot\frac{\nu-1}{\nu-2} = \frac{\nu}{\nu-2}.\end{aligned}$$

Once more, integrate y^3 times that left side by parts. For $\nu>4$ the boundary term, of order $|y|^{4-\nu}$, vanishes, and the same substitution turns the remaining integral into the second moment of $t_{\nu-2}$, $(\nu-2)/(\nu-4)$. Collecting the factors as before,

$$\begin{aligned}\mathbb{E}[Y^4] &= \frac{3\nu}{\nu-2}\cdot\frac{\nu}{\nu-2}\cdot\frac{\nu-2}{\nu-4}\\ &= \frac{3\nu^2}{(\nu-2)(\nu-4)}.\end{aligned}$$

Hence $\operatorname{Kurt}(Y)=\mathbb{E}[Y^4]/(\mathbb{E}[Y^2])^2 = 3(\nu-2)/(\nu-4) = 3+6/(\nu-4)$. For $2<\nu\le 4$ the integral of $y^4 f_Y(y)$ diverges by the power test, so the kurtosis is infinite. ■

Two fits to TAIEX

EXAMPLE 8

Two ways to fit a Student-t

Fit a Student-t to TAIEX's daily returns in two ways: by matching the mean, the standard deviation and the kurtosis, and by maximum likelihood. Compare the two fits.

SOLUTION

Kurtosis fit. LM14 computed TR23's excess kurtosis, 5.27. By (15.12), $\nu = 4 + 6/5.27 = 5.14$ (5.139 unrounded, which the exhibits use). With location $\bar{R} = 0.0625\%$ and $\sigma = s$, (15.11) gives the scale $s\sqrt{(\nu-2)/\nu} = 1.2084\% \times \sqrt{3.14/5.14} = 0.944\%$.

Maximum likelihood. SciPy's `t.fit` picks the parameters that give the returns the largest product of densities, the likelihood: this is maximum likelihood, which LM20 develops. It returns $\nu = 3.065$, location 0.0997% and scale 0.7761%. By (15.11), its standard deviation is $0.7761\% \times \sqrt{3.065/1.065} = 1.317\%$, 9% above s. The location sits near LM14's sample median, 0.0879%, rather than the mean.

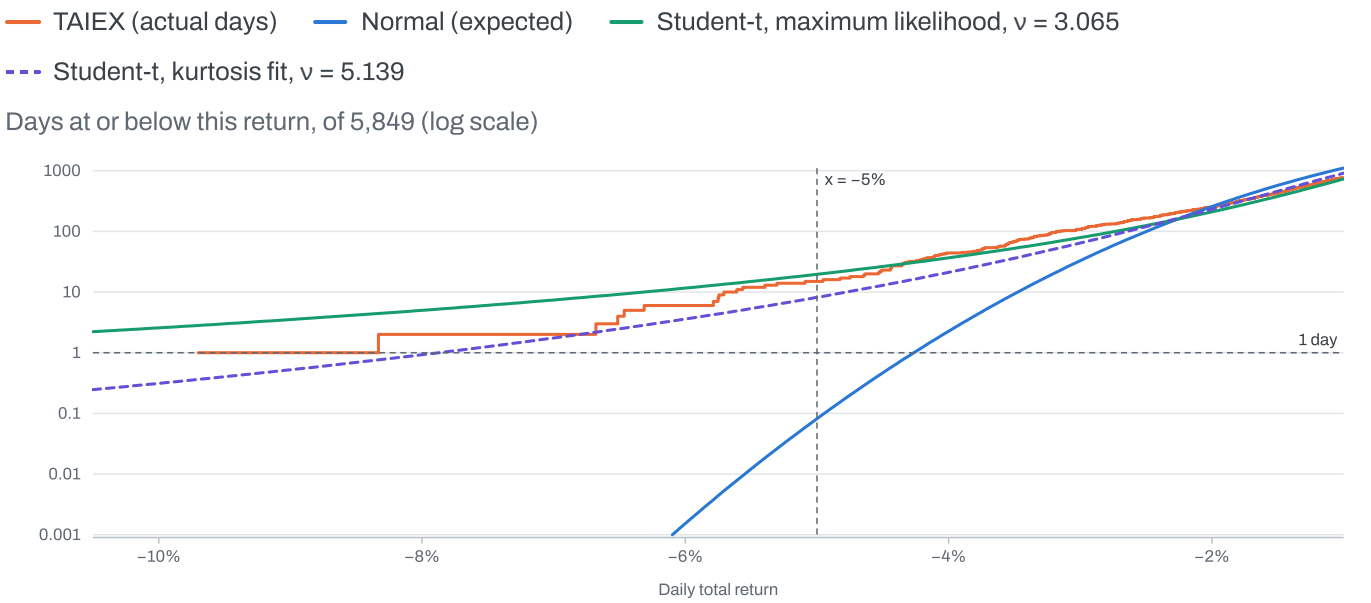
The second fit has $\nu < 4$, so the kurtosis that the first fit matched is infinite under it. At −5% the kurtosis fit expects 8.17 days and maximum likelihood 19.6, against TAIEX's 15 (Exhibit 6, Exhibit 7).

Exhibit 6: Days at or below a loss level: three models and TAIEX

Loss level x	z	Normal	Student-t, kurtosis fit	Student-t, maximum likelihood	TAIEX
2%	−1.707	257	232	210	252
3%	−2.534	32.9	64.3	79.0	108
4%	−3.362	2.26	21.2	36.6	44
5%	−4.190	0.0817	8.17	19.6	15
6%	−5.017	0.00153	3.59	11.6	6
7%	−5.845	1.5×10^{-5}	1.75	7.41	2
8%	−6.672	7.4×10^{-8}	0.926	5.00	2
9%	−7.500	1.9×10^{-10}	0.524	3.52	1
10%	−8.327	2.4×10^{-13}	0.313	2.57	0

Note: model columns are expected days, $5{,}849\,\Pr(R \le -x)$, to three significant figures (two below 0.001); TAIEX's are guarded counts (LM13 for 108, 44 and 15). The standardized level is $z = (-x - \bar{R})/s$, with $\bar{R} = 0.0625\%$ and $s = 1.20836\%$ as in the normal. The kurtosis fit has $\nu = 5.139$, location $\bar{R}$ and standard deviation s. The maximum-likelihood fit has $\nu = 3.065$, location 0.0997% and scale 0.7761%; with ν rounded to 3.07 it would expect 19.5 days at −5%. Source: as Exhibit 1.

Exhibit 7: Days at or below a return: TAIEX against three models (log scale)



Note: the number of days with R at or below each return from −10.5% to −1.0%; TAIEX's line steps once per day. The normal curve ends where it expects 0.001 days, as in LM5's chart. Models and parameters as in Exhibit 6. Source: as Exhibit 1.

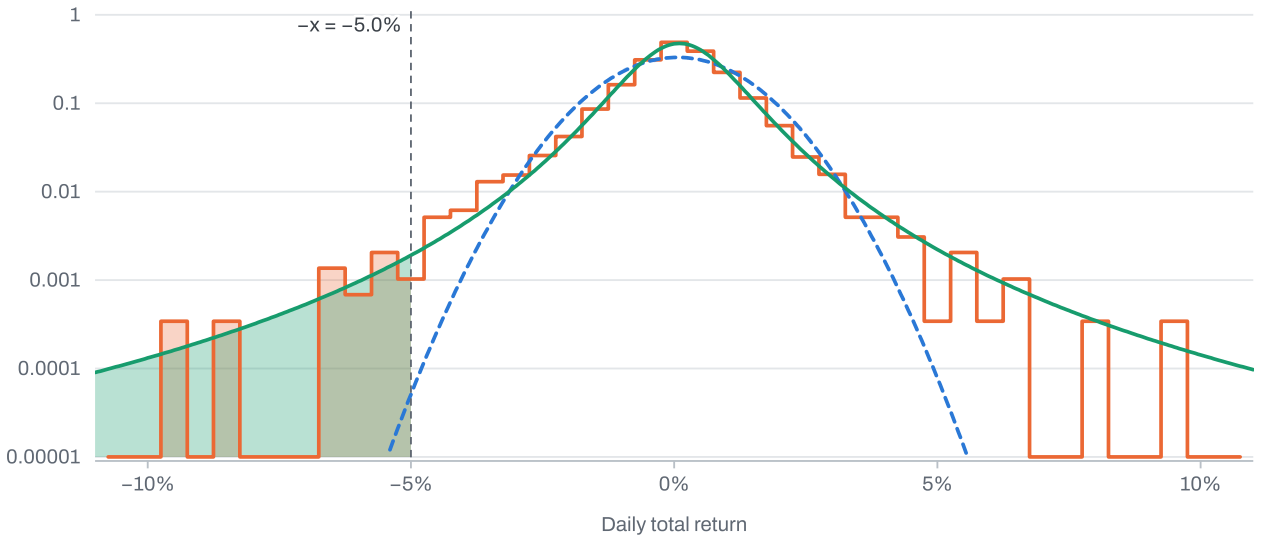
The widget draws the same models as densities beside TAIEX's histogram, for any loss level and ν.

Student-t explorer: TAIEX histogram, normal and Student-t

INTERACTIVE

TAIEX daily returns (histogram) — Normal, fitted — Student-t, maximum likelihood, ν = 3.065

Density, probability per 1 pp of return (log scale)



Standardized loss level z = (−x − mean) / s
−4.190

Normal: expected days with R ≤ −x
0.0817

Student-t: expected days with R ≤ −x
19.6

TAIEX: days with R ≤ −x
15

TAIEX ÷ Student-t (descriptive)
0.76×

Student-t: σ · kurtosis
1.317% · infinite

Student-t: 1% quantile (TAIEX −3.567%)
−3.365%

Histogram: TAIEX's 5,849 daily total returns (2003-01-03 to 2026-10-08) in 0.5-point bins, scaled to total area 1. Dashed: the normal with the data's mean and standard deviation. Solid: the Student-t, either the maximum-likelihood fit or one with the data's mean and standard deviation and the chosen ν. Shaded: R ≤ −x. Every fit is in-sample; TAIEX's days are counted with the guard R ≤ −x + 10^{-12}.

PITFALL **Matching a moment the fit says does not exist**

Formula (15.12) presumes $\nu > 4$, so that the kurtosis exists. The maximum-likelihood fit puts ν at 3.065, where the population kurtosis is infinite and a sample kurtosis estimates nothing stable (Section 7). Read SciPy's parameters with care too: its `scale` is not σ, which by (15.11) is scale $\times \sqrt{\nu/(\nu-2)}$.

導讀 **t 分布：自由度決定尾巴**

t 分布（Student-t distribution）的密度（density）在尾端以 $|y|^{-(\nu+1)}$ 衰減，只有階數小於自由度（degrees of freedom）ν 的動差（moment）存在：$\nu > 2$ 時變異數（variance）是 $\nu/(\nu-2)$，$\nu > 4$ 時峰態（kurtosis）是 $3 + 6/(\nu-4)$，而 $2 < \nu \le 4$ 時峰態是無窮大。密度前面的常數由伽瑪函數（gamma function）Γ 給出，$\nu = 3$ 時是 $2/(\pi\sqrt{3})$，正是 LM5 省略的那個常數；ν 趨於無窮大時，t 分布趨近標準常態分布（standard normal distribution）。由加權股價報酬指數的超額峰態（excess kurtosis）5.27 反推，得到 $\nu = 5.14$，用 maximum likelihood 估計卻得到 $\nu = 3.065$，在這個參數下峰態根本不是有限值。在 −5% 以下，常態模型預期 0.0817 天，maximum likelihood 的 t 分布預期 19.6 天，這個指數實際是 15 天。

KNOWLEDGE CHECK 5

Which moments of t_ν exist for $\nu = 3.065$, the maximum-likelihood fit, and for $\nu = 5.139$, the kurtosis fit?

ANSWER

The moments of order below ν exist. At $\nu = 3.065$ the mean, the variance and the third moment exist, but the kurtosis is infinite. At $\nu = 5.139$ the moments up to order 5 exist, and by (15.10) the kurtosis is $3 + 6/1.139 = 8.27$: an excess kurtosis of 5.27, as intended.

7 TAIL MEASURES AND CHOOSING A MODEL

- ☐ compare tail models through exceedance counts and quantiles, and read a tail index from log–log chords
- ☐ choose a distribution for a stated question and name the assumption that limits it

A kurtosis that does not settle

LM5 warned that a sample kurtosis never settles when the tail index is near 3, and advised tail measures that need fewer moments. TR23 shows both points, but a deep chord rests on a few days and moves too.

EXAMPLE 9

One day, four tail measures

Compute four tail measures on the 5,478 returns to 2 April 2025 and on the 5,479 returns to 7 April 2025. The measures are the excess kurtosis, the 1% quantile, the count of days at or below −3% and the chord (15.13) from −5% to −6%, defined below. Then remove 7 April 2025, and the four largest moves, from the full sample.

SOLUTION

In one day the excess kurtosis rose from 4.25 to 4.94. The 1% quantile moved from −3.538% to −3.543%, and the count of days at or below −3% from 96 to 97. The days at or below −5% and −6% went from 12 and 4 to 13 and 5. One return moved the kurtosis by 0.69, the quantile by 0.005 points, the count by one day and the chord by 0.79, from 6.03 to 5.24.

Without 7 April 2025, the full sample's excess kurtosis falls from 5.27 to 4.71. Without the four largest moves (7 and 10 April 2025, 5 August 2024 and 31 July 2026) it falls to 3.74. The ten largest fourth-power terms supply 35.0% of $\sum_t (R_t - \bar{R})^4$ but only 6.7% of $\sum_t (R_t - \bar{R})^2$.

LM5's 0050 sample behaved alike: 8.12, then 7.05 and 3.85 after the same removals, 5.84 to 7.63 in one day, and shares of 62.8% and 14.1%.

Exhibit 8: Tail measures in the two limit regimes

	Full sample	7% limit	10% limit
Dates	3 Jan 2003 – 8 Oct 2026	3 Jan 2003 – 29 May 2015	1 Jun 2015 – 8 Oct 2026
Returns T	5,849	3,077	2,772
s (%)	1.208	1.257	1.152
Excess kurtosis	5.27	3.39	8.07
ν by kurtosis, (15.12)	5.14	5.77	4.74
ν by maximum likelihood	3.065	2.990	3.237
Rank of the 1% quantile, $\lceil 0.01T \rceil$	59	31	28
1% quantile (%)	−3.567	−3.621	−3.467
Days with $R \le -3\%$ per 1,000	18.5	23.4	13.0
Monthly counts of $\lvert R \rvert \ge 3\%$: variance ÷ mean	3.02	2.95	3.01

Note: a stability check; each part is fitted on its own. A part's quantile is LM14's order statistic with the part's size. The parts' months are January 2003 – May 2015 (149) and June 2015 – October 2026 (137). No standard errors are given (LM20); 5.27 is LM14's value. Source: as Exhibit 1.

Exhibit 8 repeats the measures in the two limit regimes: the excess kurtosis moves most, from 3.39 to 8.07 (Problem 11). LM17 follows the running sample kurtosis day by day.

Chords of a power tail

Counts and quantiles need no moments. Recall LM5's power tail: a probability that falls like $x^{-\alpha}$ is a line of slope $-\alpha$ on log–log axes.

PROPOSITION **Tail index from two levels**

Let $0 < x_1 < x_2$, and suppose that $\Pr(R \le -x)$ is a constant multiple of $x^{-\alpha}$ for $x_1 \le x \le x_2$. Then

$$\alpha = \frac{\ln\left[\Pr(R \le -x_1)/\Pr(R \le -x_2)\right]}{\ln(x_2/x_1)}. \tag{15.13}$$

For any other law, the right side, the **chord** between the two levels, is a reading of the tail, not a parameter.

Proof. The constant cancels in the ratio, which equals $(x_2/x_1)^{\alpha}$; take logarithms. ■

EXAMPLE 10

Reading a tail index

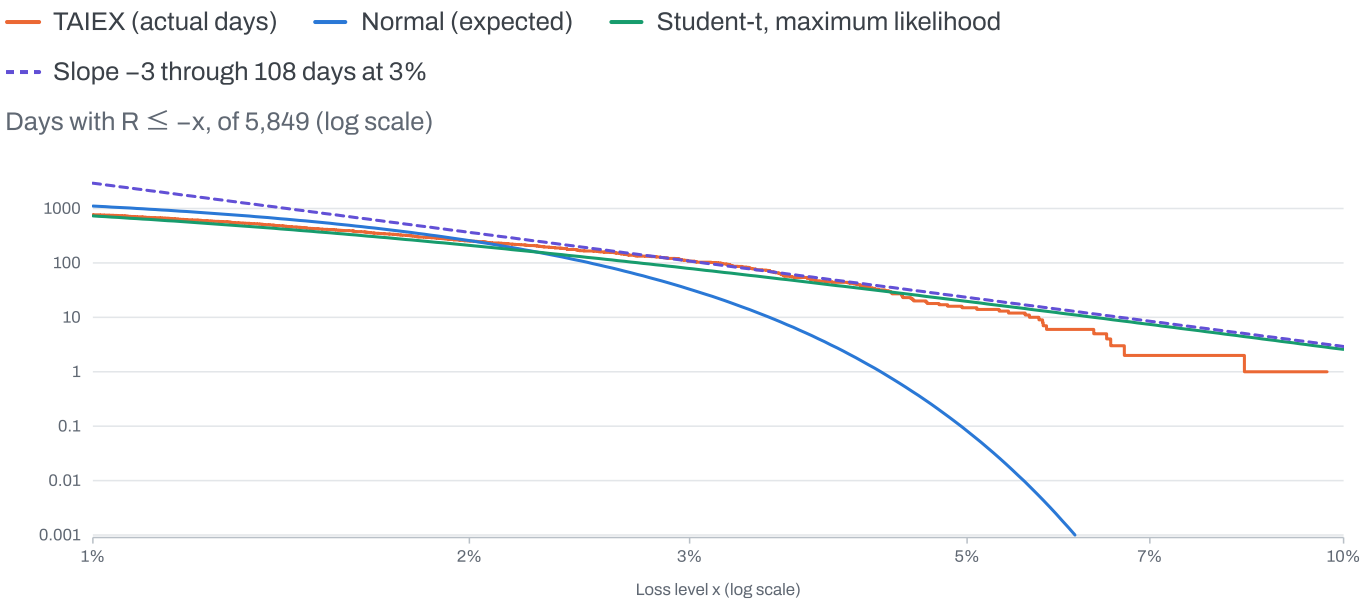
Use Exhibit 6's counts at −2% to −6% to compute the chords (15.13) between adjacent levels. Compare the chord from −3% to −6% with those of the three models.

SOLUTION

The counts 252, 108, 44, 15 and 6 give, for example, $\ln(252/108)/\ln 1.5 = 2.09$ from −2% to −3%. The adjacent chords are 2.09, 3.12, 4.82 and 5.03: the chord rises with the level (Exhibit 9). All ten pairs of levels from −2% to −6% give chords between 2.09 and 5.03, mostly inside LM5's range of 2 to 5.

From −3% to −6% the data give $\ln(108/6)/\ln 2 = 4.17$. The normal's chord is 14.4, the maximum-likelihood fit's 2.77 and the kurtosis fit's 4.16. Report the chords, not one value: LM20 estimates a tail index with its standard error.

Exhibit 9: Days at or below a loss level on log–log axes



Note: days with $R \leq -x$ for x from 1% to 10%, both axes logarithmic. The dashed line has slope −3 and passes through TAIEX's 108 days at 3%: LM5's tail index of about 3. The adjacent chords, 2.09, 3.12, 4.82 and 5.03 from −2% to −6%, rise with the level. The steepening beyond 5% rests on few days (15, 6, 2, 2 and 1 at 5% to 9%). Before June 2015 the 7% limit kept every return above −6.68%. Source: as Exhibit 1.

Large falls outnumber large rises, 108 against 67; LM14 computed TR23's skewness, −0.34, and LM19 studies it as a statistic. Extreme value theory models the largest losses directly; it is beyond this book.

> **PITFALL** **The price limit truncates the tail**
>
> TWSE caps each stock's daily move, at 7% until 29 May 2015 and at 10% from 1 June 2015, so the index cannot fall by more in one session. Before June 2015 every TR23 return lay between −6.6789% and +6.7420%. The sample's worst day, −9.6997% on 7 April 2025, was capped for many stocks (LM14). A model without bounds spreads mass beyond the limit: compare models inside it.

Choosing a model

Exhibit 10 resolves the case: the question decides the model, and each model has an assumption that limits it.

Exhibit 10: Which model for which question

Question	Model	Evidence	Limiting assumption
Days at or below −2%	normal	257 days against 252	thin tails: fails from −3%
The 5% quantile	normal	−1.925% against −1.862%	thin tails
Days from −3% to −5%; the 1% quantile	Student-t, maximum likelihood	79.0, 36.6, 19.6 days against 108, 44, 15; −3.365% against −3.567%	symmetry; too heavy from −5%
Days beyond −7%	counts	2, 2, 1 days at −7%, −8%, −9%; the kurtosis fit's 1.75, 0.926, 0.524 come close, but its ν rests on the unstable kurtosis	few days, cut by the price limit
A calendar year's growth	lognormal	losing years 6.90 expected against 5; beyond the 5% and 95% quantiles 2.3 against 2; mean +15.16% against 14.97%	independent years
Large moves in a month	none: not the Poisson	variance 1.846 against mean 0.612	clustering
Up days in 20 sessions	binomial	variance 4.72 against 4.95	independent days

Note: evidence from Exhibit 1, Example 1, Example 4, Exhibit 6, Example 6 and Exhibit 5; every model is fitted in-sample. The lognormal row uses the calendar years 2003–2025, 2003 from its first close. The maximum-likelihood Student-t's 1% quantile is $0.0997\% + 0.7761\% \times F_Y^{-1}(0.01)$ with $\nu = 3.065$.

Tail means rank the models the same way. The 15 days at or below −5% averaged −6.26%. The normal's (5.11) gives −5.26%, and the maximum-likelihood Student-t gives −7.57%. With its sign reversed, the mean return beyond a quantile is the expected shortfall of LM32.

導讀 **尾部：少用高階動差**

樣本峰態（sample kurtosis）由少數極端日主導：2025 年 4 月 7 日一天就讓加權股價報酬指數的超額峰態（excess kurtosis）從 4.25 跳到 4.94，同一天 1% 分位數（quantile）只移動 0.005 個百分點，−3% 以下的天數只多 1 天。最大的十個四次方項占全部的 35.0%，拿掉四個最大的漲跌日，超額峰態就降到 3.74。描述尾巴應優先使用需要較少動差（moment）的量：超過門檻的天數、分位數，以及雙對數圖（log–log plot）上相鄰門檻之間的弦（chord）斜率，也就是尾部指數（tail index）的讀數；加權股價報酬指數的弦在 2.09 到 5.03 之間，門檻越深越陡，但深處的弦只靠少數幾天，同樣不穩。沒有一個分布能回答所有問題：−2% 用常態分布（normal distribution），−3% 到 −5% 用 maximum likelihood 的 t 分布（Student-t distribution），年度成長用對數常態分布（lognormal distribution），每個模型都有限制它的假設。

KNOWLEDGE CHECK 6

The maximum-likelihood Student-t expects 2.57 days at or below −10%, and TAIEX has none. Is the fit wrong there?

ANSWER

Yes, but the market rule shows it, not the count. Each stock's daily move is capped at 10% (7% before June 2015), so the index can fall 10% only if nearly every stock closes at its limit. A model without bounds, like the Student-t, spreads probability beyond the cap: compare models inside it.

8 DISTRIBUTIONS IN PYTHON

- [ ] implement the fits and counts with `scipy.stats` and the counting guard

SciPy's distributions share `pdf`, `cdf`, `ppf`, `sf` and `fit`. The calls of this module:

PYTHON

```python
import numpy as np
from scipy import stats
# R: TR23's 5,849 daily returns as a NumPy array
T = len(R)
down5 = int((R <= -0.05 + 1e-12).sum())        # 15 days: the guard
loc, scale = stats.norm.fit(R)                  # mean and s, divisor T
nu, loc_t, sc_t = stats.t.fit(R)                # maximum likelihood
print(T * stats.norm.cdf(-0.05, loc, scale))    # 0.0817 days
print(T * stats.t.cdf(-0.05, nu, loc_t, sc_t))  # 19.6 days
print(stats.t.ppf(0.01, nu, loc_t, sc_t))       # -0.03365
print(np.quantile(R, 0.01, method="inverted_cdf"))  # -0.03567
print(stats.kurtosis(R))                        # 5.27, excess kurtosis
lg = stats.lognorm(s=0.22207, scale=np.exp(0.11646))
print(lg.median(), lg.mean(), lg.ppf([0.05, 0.95]))
print(286 * stats.poisson(mu=175 / 286).pmf(0))  # 155.1 months
print(292 * stats.binom(20, 0.548).cdf(5))       # 1.98 blocks
```

Exhibit 11 maps SciPy's parameters to this module's symbols. LM18 draws from these laws.

Exhibit 11: SciPy's distributions in this module's symbols

Call	Parameters	Trap
`norm(loc, scale)`	μ, σ	`norm.fit` returns $\bar{R}$ and s, divisor T
`lognorm(s, scale)`	σ_X, e^{μ_X}	`s` is σ_X, not a standard deviation of G
`t(df, loc, scale)`	ν, location, $\sigma\sqrt{(\nu-2)/\nu}$	`scale` is not σ; the variance is `scale**2*df/(df-2)` for `df > 2`
`poisson(mu)`	λ	the parameter is named `mu`
`binom(n, p)`	n, p	`cdf(5)` includes 5
`.fit(data)`	maximum likelihood	the default; `method="MM"` matches the first three raw moments of `t` , one per free parameter, not (15.12)'s kurtosis; fails on TR23
`stats.kurtosis(R)` , `.stats(moments="k")`	excess kurtosis	`stats.kurtosis` uses divisor-T moments, `.stats` the model's; `kurtosis(R, fisher=False)` returns Kurt
`np.quantile(R, u, method="inverted_cdf")`	$\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$	the default method interpolates

Note: SciPy 1.18 and NumPy 2.5 references; the divisor of `norm.fit` and the variance of `t` were checked by computation (`data/lm15_numbers.py`).

PITFALL **Excess or not**

SciPy's `stats.kurtosis` returns the excess kurtosis Kurt -3 by default (`fisher=True`), and a distribution's `.stats(moments="k")` does too. LM5's tails exhibit and (15.10) give the kurtosis itself.

IN PYTHON **Companion lab**

The Polars-first notebook `LM15_lab.ipynb` rebuilds the module's exhibits and computational examples, fits the Student-t from several starting points, and solves the computational practice problems. It ends with a Student-t fitted to five-session log returns.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py` , then `uv run python data/make_lab_data.py` . FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

SciPy prints 6 for the kurtosis of t_5, while LM5's tails exhibit lists 9. Which is right?

ANSWER

Both. By (15.10), $\text{Kurt}(Y) = 3 + 6/(5-4) = 9$, the figure LM5 lists. SciPy's `t(5).stats(moments="k")` reports the excess kurtosis, $9 - 3 = 6$.

SUMMARY

Question	Distribution	Result in this module
How many successes in n trials?	binomial (15.1)–(15.2)	up days in 20 sessions: variance 4.72 against 4.95
How many rare events?	Poisson (15.3)–(15.5)	large moves a month: variance 1.846 against mean 0.612
Which loss on 1% of days?	normal quantile (15.6)	−2.749% against TAIEX's −3.567%
How does a year grow?	lognormal (15.7)	median 1.1235, mean 1.1516, losing year 30.0%
How heavy is the tail?	Student-t (15.9)–(15.12); chords (15.13)	$\nu = 3.065$; 19.6 days at −5% against 15

- The binomial fits up days in blocks, which is consistent with independent days but does not prove it.
- Rare binomial trials tend to the Poisson. Monthly counts of large moves are overdispersed: large moves cluster.
- TAIEX's QQ plot bends away from the normal's line in both tails.
- A lognormal's median is e^{μ_X}; fitted to calendar years, it is the compound rate.
- The Student-t's kurtosis needs $\nu > 4$, and maximum likelihood puts TAIEX at $\nu = 3.065$.
- A sample kurtosis rests on a few days; counts, quantiles and chords need fewer moments.
- Choose a distribution per question, and name the assumption that limits it.

PRACTICE PROBLEMS

Every problem uses TR23: TAIEX's daily total returns, 3 January 2003 – 8 October 2026, $T = 5{,}849$, $A = 246.12$. Moments divide by T, counts use the guard $R \le -x + 10^{-12}$, and an empirical quantile is LM14's $R_{(\lceil uT \rceil)}$ (`method="inverted_cdf"`; in a part, the part's size replaces T).

1. Up days in a 20-session block are modelled as $\mathrm{Bin}(20, 0.548)$. The variance of the count is *closest* to:

   A. 4.95

   B. 10.96

   C. 2.23

2. Let $X \sim \mathrm{Poisson}(0.612)$. Compute $\Pr(X = 0)$ and $\Pr(X \ge 5)$ to three significant figures, and the expected numbers of such months among 286, against the 210 and 9 observed. Which months does the model miss, and why?

3. The 286 monthly counts of days with $|R| \ge 3\%$ have mean 0.612 and variance 1.846 (divisor 286). Their ratio of variance to mean is *closest* to:

   A. 0.33

   B. 1.00

   C. 3.02

4. For $X \sim \mathrm{Poisson}(\lambda)$, derive $\mathbb{E}[X(X-1)] = \lambda^2$ from (3.2), and deduce $\mathrm{Var}(X) = \lambda$. Then show that $\mathrm{Bin}(n, \lambda/n)$ has variance $\lambda(1 - \lambda/n)$, which tends to λ.

5. With $\bar{R} = 0.0625\%$, $s = 1.2084\%$ and $\Phi^{-1}(0.99) = 2.3263$, compute the normal's 1% and 99% quantiles to three decimals. TAIEX's are $R_{(59)} = -3.567\%$ and $R_{(5{,}791)} = +3.125\%$. In which tail does the normal miss more, and by how many points does it miss each tail?

6. The lognormal fitted to the 23 calendar-year growth factors 2003–2025, 2003 from its first close, has $\mu_X = 0.11646$ and $\sigma_X = 0.22207$. The probability it gives a losing year is *closest* to:

A. 21.7%

B. 30.0%

C. 70.0%

7. For the lognormal of Problem 6, compute the median, the mean and the standard deviation to four decimals, and the 5% and 95% quantiles with $\Phi^{-1}(0.95) = 1.6449$. Check them with `stats.lognorm(s=0.22207, scale=np.exp(0.11646))`.

8. Show that the constant of (15.9) at $\nu = 4$ is $\Gamma(\frac{5}{2})/\left(\sqrt{4\pi}\,\Gamma(2)\right) = 3/8$. Give the variance and kurtosis of t_5 and t_6. Which of the two exist for t_4, and which for t_2?

9. TR23's excess kurtosis is 5.27. The ν of a Student-t with the same kurtosis is *closest* to:

   A. 1.14

   B. 6.64

   C. 5.14

10. TAIEX has 252, 44 and 6 days at or below −2%, −4% and −6%. Compute the chords (15.13) from −2% to −4%, from −4% to −6% and from −2% to −6% (two decimals). Why do they differ, and how do they compare with the maximum-likelihood fit's chords at the same levels and its $\nu = 3.065$?

11. *(Python, stability check)* Split TR23 at 1 June 2015. For each part, compute T, s (in %, three decimals), the excess kurtosis (two decimals) and the ν of `stats.t.fit` (three decimals). Add the 1% quantile: $R_{(31)}$ of the 3,077 returns before the split, $R_{(28)}$ of the 2,772 after. Which measure moved most, and which label does the comparison carry?

12. *(Python)* Count the days at or above +3% and +5%. Compare them with the expected days of the normal and of the maximum-likelihood Student-t (three significant figures), and with the lower tail's 108 and 15. Why can no symmetric model centred near the median fit both tails?

SOLUTIONS

1. **A** is correct: by (15.2), $20 \times 0.548 \times 0.452 = 4.95$. B, 10.96, is the mean np. C, 2.23, is the standard deviation.

2. By (15.3), $\Pr(X = 0) = e^{-0.612} = 0.542$ and $\Pr(X \geq 5) = 1 - \sum_{j \leq 4} \Pr(X = j) = 0.000431$. Of 286 months, the model expects $286e^{-0.612} = 155.1$ and $286 \times 0.000431 = 0.123$ months. The model misses both ends, 210 calm and 9 turbulent months. Large moves cluster because volatility changes over time; the Poisson assumes a constant rate.

3. **C** is correct: $1.846/0.612 = 3.02$. A, 0.33, inverts the ratio. B, 1.00, is a Poisson count's ratio; the data are overdispersed.

4. Drop the terms $j = 0$ and 1 and shift the index by 2: $\mathbb{E}[X(X-1)] = \lambda^2 \sum_{j \geq 2} e^{-\lambda}\lambda^{j-2}/(j-2)! = \lambda^2$, by (3.2). Then $\mathrm{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - (\mathbb{E}[X])^2 = \lambda$. By (15.2), $\mathrm{Bin}(n, \lambda/n)$ has variance $n \cdot \frac{\lambda}{n}\left(1 - \frac{\lambda}{n}\right) = \lambda(1 - \lambda/n)$, which tends to λ.

5. By (15.6), the 1% quantile is $0.0625\% - 1.2084\% \times 2.3263 = -2.749\%$ and the 99% quantile $+2.874\%$. The normal misses the lower tail more: by 0.82 points, against 0.25 in the upper tail. Large falls are larger than large rises: TR23's skewness is −0.34 (LM14).

6. **B** is correct: $\Phi(-0.11646/0.22207) = \Phi(-0.5244) = 30.0\%$. A, 21.7%, is the sample's 5 losing years of 23, not the model's probability. C, 70.0%, reverses the sign: it is $\Pr(G > 1)$.

7. By (15.7), the median is $e^{0.11646} = 1.1235$, the mean $e^{0.11646 + 0.22207^2/2} = 1.1516$ and the standard deviation 0.2589. The quantiles are $e^{0.11646 \mp 1.6449 \times 0.22207} = 0.7797$ and 1.6189. SciPy's `median()`, `mean()`, `std()` and `ppf([0.05, 0.95])` return the same values.

8. With $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$ and $\Gamma(2) = 1$, the constant is $\frac{3}{4}\sqrt{\pi}/(2\sqrt{\pi}) = 3/8$. By (15.10), t_5 has variance $5/3$ and kurtosis 9, and t_6 has variance $3/2$ and kurtosis 6. For t_4 the variance, 2, exists, but the kurtosis is infinite. For t_2 the variance is infinite and the kurtosis undefined.

9. **C** is correct: by (15.12), $\nu = 4 + 6/5.27 = 5.14$. A, 1.14, forgets the 4. B, 6.64, takes the excess kurtosis for the kurtosis, $4 + 6/(5.27 - 3)$.

10. The chords are $\ln(252/44)/\ln 2 = 2.52$, $\ln(44/6)/\ln 1.5 = 4.91$ and $\ln(252/6)/\ln 3 = 3.40$, the mean of the first two weighted by $\ln 2$ and $\ln 1.5$. In TR23 the chord rises with the level, so one power law does not describe the counts from −2% to −6%. The deeper chord rests on 6 days and on the price limit (Exhibit 9).

    Far in its tail a Student-t's probability falls like $x^{-\nu}$, so 3.065 is that model's tail index. At these levels the fit's own chords, from Exhibit 6's 210, 36.6 and 11.6 days, are 2.52, 2.83 and 2.64. They equal the data's from −2% to −4% and lie far below them from −4% to −6%.

11. Before June 2015: $T = 3{,}077$, $s = 1.257\%$, excess kurtosis 3.39, $\nu = 2.990$, 1% quantile −3.621%. From June 2015: $T = 2{,}772$, $s = 1.152\%$, excess kurtosis 8.07, $\nu = 3.237$, 1% quantile −3.467%. The excess kurtosis moved most, more than doubling (Exhibit 8). The comparison is a stability check: each part is fitted in-sample, and no standard error says whether a change exceeds chance (LM20, LM21).

12. TAIEX has 67 days at or above +3% and 13 at or above +5%. The normal expects 44.0 and 0.128. The maximum-likelihood Student-t expects 94.2 and 22.0, so it overstates the upper tail at both levels. In the lower tail it understates at −3% (79.0 against 108) and overstates at −5% (19.6 against 15).

    A symmetric model mirrors its tails about its centre. Centred at +0.0997%, the Student-t even expects more large rises than large falls, while TAIEX has 108 large falls and 67 large rises.

GLOSSARY

Term	中文	Meaning
Bernoulli distribution	柏努利分布	Law of one trial: success with probability p; mean p, variance $p(1-p)$.
Binomial distribution	二項分布	Number of successes in n independent trials with one p, (15.1); mean np, variance $np(1-p)$.
Poisson distribution	卜瓦松分布	Law of a count with $\Pr(X=j)=e^{-\lambda}\lambda^j/j!$, (15.3); mean and variance λ.
Overdispersion	過度離散	A count's variance above its mean: a sign that a Poisson does not fit.
Independent and identically distributed (iid)	獨立同分布	Independent random variables that share one distribution.
Quantile function	分位數函數	$F_X^{-1}(u)$, the smallest value at which F_X reaches u; for a normal, $\mu+\sigma\Phi^{-1}(u)$, (15.6).
QQ plot	分位數-分位數點圖	Sorted standardized returns against normal quantiles at the levels $(j-\frac{1}{2})/T$.
Empirical distribution	經驗分布	Mass $1/T$ on each observation; its quantiles are order statistics.
Lognormal distribution	對數常態分布	Law of $G=e^X$ for a normal X, (15.7); median e^{μ_X}, mean $e^{\mu_X+\sigma_X^2/2}$.
Gamma function	伽瑪函數	$\Gamma(a)=\int_0^\infty y^{a-1}e^{-y}dy$, (15.8), with $\Gamma(a+1)=a\Gamma(a)$.
Student-t distribution	t 分布	Symmetric law with density (15.9) and power tails; tends to the normal as $\nu\to\infty$.
Degrees of freedom	自由度	The Student-t's parameter ν: moments of order below ν exist.

Term	中文	Meaning
Excess kurtosis	超額峰態	Kurt -3; 0 for a normal; what SciPy's `kurtosis` returns.
Heavy tail	厚尾	A tail that falls like a power of the loss rather than like the normal's $e^{-z^2/2}$.
Tail index	尾部指數	The exponent α of a power tail; between two levels, read as a chord (15.13).

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